

Q5: Let x and y are two independent r.v.'s with joint pdf.

$$f(x, y) = \frac{x(1 + 3y^2)}{4}, \quad 0 < x < 2, 0 < y < 1$$

$$= 0, \quad \text{elsewhere.}$$

a. Show that marginal pdf of X is

$$g(x) = \frac{x}{2} ; \text{ for } 0 < x < 2$$

b. Also calculate $E(X)$.

Q5: a. If approximately 2% of the people in a room of 200 people are left-handed, find the probability that exactly 5 people there are left-handed. Use Poisson distribution.

b. A batch of 10 rocker cover gaskets contains 4 defective gaskets. If we draw samples of size 3 without replacement, from the batch of 10, find the probability that a sample contains 2 defective gaskets.

Q5: A random sample of 50 observations is drawn from a normal distribution with $\sigma^2 = 52$.

Suppose that the sample mean is found to be $\bar{X} = 71$. Find 95% confidence interval for μ . (Use table value as $Z_{\alpha/2} = 1.96$)

Q5: From the paired samples from normal population we have $n = 25$, $\bar{d} = 3$, $s_d = 2.5$ and t-table

$$\text{If } f(x) = \frac{1}{5}x, f(y) = 2y, \text{ and } f(x, y) = \frac{2}{5}xy$$

Then prove that x and y are independent.

Q3: If random samples of size three is drawn without replacement from the population consisting of four numbers 4, 5, 5, 7. Draw all possible samples and sample means.

Q3: If $\bar{X}_1 - \bar{X}_2 = 24, \sigma_1^2 / n_1 = 15, \sigma_2^2 / n_2 = 10, \alpha = 0.10, H_0 : \mu_1 - \mu_2 = 22$

Find the value of z-test statistic.

Solution:

Q:3 Elaborate the Least Significant Difference (LSD) test.

If X and Y are independent variables, and $E(X) = 3.2$, $E(Y) = 3.0$, then find $E(XY)$.
Q2:

Q2: Suppose we are interested to know the sex ratio of females in a city. In this situation which type of probability distribution we can use?

Q2: If $n=9$ and $\alpha = 5\%$ then find the value of $t_{\alpha/2(n-1)}$

Q2: Briefly discuss about pairing by design in t-distribution?